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**To:** The Editor-in-Chief, *IEEE Signal Processing Letters*

**From:** Bin Nong, Weihong Fu, and Zhuoyun Jiang

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**Manuscript title:** A Label-Score Alignment Pitfall in SNR-Conditioned OOD Detection for Single-Channel Blind Source Separation

**Manuscript type:** Letter (original submission)

Dear Editor,

We submit this Letter reporting an evaluation-methodology pitfall in SNR-conditioned out-of-distribution (OOD) detection for single-channel blind source separation (SC-BSS) of co-frequency communication signals. In our own research pipeline, two deterministic label-score misalignments—neither detectable by multi-seed averaging, significance tests, or robustness batteries—simultaneously manufactured a spurious SNR-routing gain (AUROC  $0.50 \rightarrow 0.625$ ) and erased a genuine same-mixture detection margin ( $0.54 \rightarrow \text{chance}$ ). We formalize the failure as an indexing-permutation confound (Proposition 1 with a closed-form Gaussian corollary) and show with a domain-free analytical toy that it needs no separator, signal, or detector: it is a general confound of condition-conditioned evaluation, directly relevant to the many signal-processing studies that report per-SNR or per-condition metrics. The Letter closes with a minimal audit protocol that catches both failure modes at zero cost.

The manuscript is original, has not been published previously, and is not under consideration by any other journal. No part of it has been posted as a preprint. All authors have approved the submission. Code, score dumps, and a one-click label-audit script reproducing every reported number are publicly available (link in the manuscript).

We hope you find the Letter suitable for *IEEE Signal Processing Letters*.

Sincerely,

Bin Nong, on behalf of all authors